DATA SCIENCE CAPSTONE

Falcon 9 first-stage landing prediction

Data Science Capstone Project Report | Historical IBM Skills Network dataset

Can a Falcon 9 booster land?

Exploratory analysis, SQL, interactive maps and a comparison of four classification models.

90

Launches analyzed

60

Successful landings

77.8%

Holdout model accuracy

Project repository: https://github.com/cpp5144/spacex-falcon9-landing-capstone

OVERVIEW

Executive summary

- The released historical dataset has 90 Falcon 9 launches from 2010 through 2020. The first stage landed successfully in 60 cases (66.7%).
- Landing rates vary by launch site: 60.0% at CCAFS SLC 40, 77.3% at KSC LC 39A and 76.9% at VAFB SLC 4E. These are descriptive associations, not causal effects.
- A shallow decision tree won five-fold cross-validation on the training split (83.4% mean accuracy). Its untouched 18-launch test split scored 77.8%, versus 66.7% for always predicting success.

INTRODUCTION

Project question and context

- SpaceX reuses first-stage boosters. Predicting whether a stage will land can help describe mission risk in this historical teaching dataset.
- Question: Given flight and booster attributes available in the course data, can a classifier distinguish successful first-stage landings from other outcomes?
- The analysis combines exploratory plots, ten SQL questions, location and dashboard views, and a held-out model comparison.

Outcome definition: Class = 1 for a successful first-stage landing; Class = 0 otherwise.

METHODOLOGY

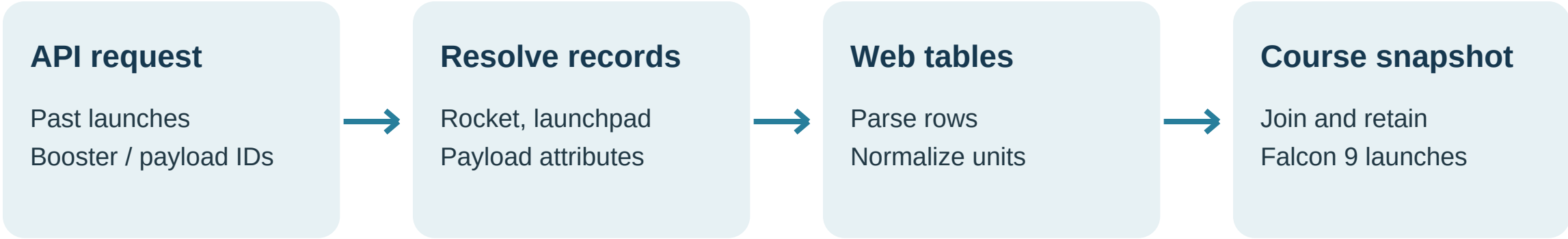
Data sources and collection

- IBM Skills Network released dataset_part_2.csv (90 labeled launches) and dataset_part_3.csv (83 model features), aligned by FlightNumber.
- The separate Spacex.csv contains 101 mission records for the SQL lab. The dashboard CSV contains 56 rows, so dashboard counts should not be compared directly with the 90-row analysis.
- The course collection path queries the SpaceX REST API for launch records and scrapes web tables for supplementary fields. This report reproduces analysis from the released IBM snapshots rather than claiming a live API extract.

METHODOLOGY

API and scraping workflow

Collection method taught in the course



Reproduction input: official IBM course CSVs. Live collection code is not claimed as executed.

METHODOLOGY

Data wrangling

- Parsed Date to year for trend analysis, retained IBM's binary Class label, and verified that the 90 feature rows align with the 90 labeled rows by FlightNumber.
- LandingPad is missing in 26 of 90 rows. It is not used directly as a numeric modeling input; the course's encoded feature table supplies the model matrix.
- Converted categorical orbit, site, serial and hardware fields into one-hot indicators in the provided feature table. Standardization is fitted on training folds only to avoid test leakage.

METHODOLOGY

EDA and SQL approach

- Scatter plots compare flight number and payload mass with launch site and orbit; point color shows the landing outcome.
- A bar chart compares landing rates by orbit, and a line chart tracks the annual mean. Small orbit groups are shown but interpreted cautiously.
- Ten SQL queries answer the course's launch site, payload, booster, mission outcome and date questions using the separate 101-row IBM SQL snapshot.

Interactive analytics

- Folium maps use recorded latitude and longitude to mark the three sites and show launch count and landing rate in each marker's popup.
- A local Plotly dashboard filters the 56-row dashboard dataset by launch site and payload range. It updates a success pie chart and a payload-versus-outcome scatter plot.
- The local HTML map and dashboard remain interactive; the report includes static summaries of their outputs.

METHODOLOGY

Classification design

- A stratified 80/20 split leaves 72 launches for training and 18 for the untouched test set; random seed 2 fixes the split.
- Within training data, five-fold cross-validation selects hyperparameters for logistic regression, a linear SVM, k nearest neighbors and a shallow decision tree.
- For each fold, the feature mean and standard deviation come from that fold's training partition. The held-out test set is evaluated once after selection.

With only 90 records, accuracy estimates are sensitive to the split and should not be treated as operational forecasts.

EDA RESULTS

Flight number and launch site

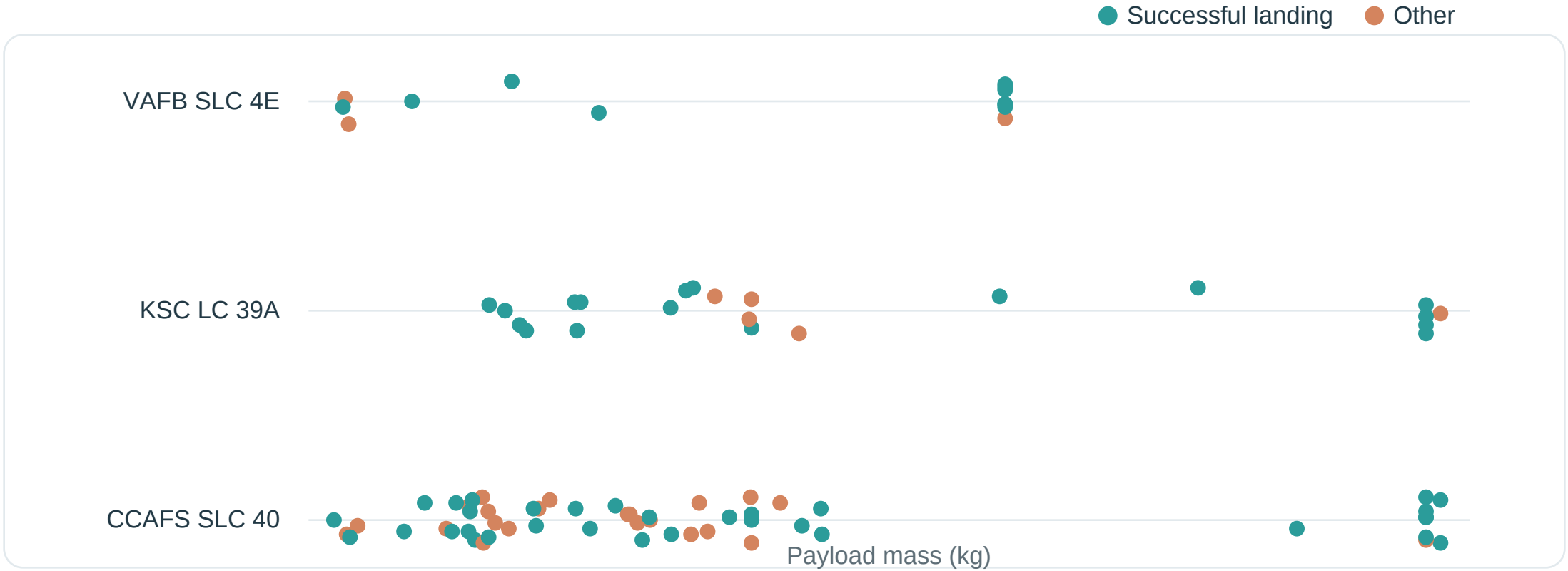
Later flights cluster at KSC and VAFB; teal points denote successful landings.



EDA RESULTS

Payload mass and launch site

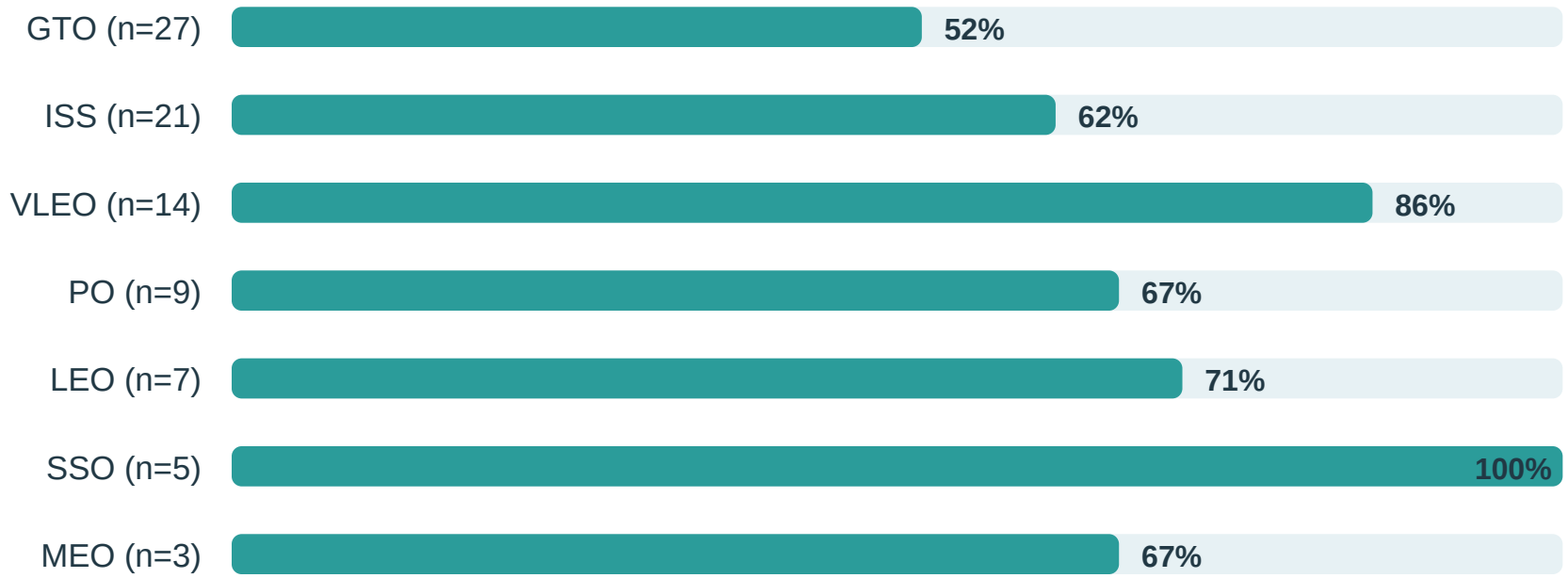
The sites serve different payload ranges; outcomes overlap within each site.



EDA RESULTS

Landing rate by orbit

Orbit categories with few observations have unstable rates.



EDA RESULTS

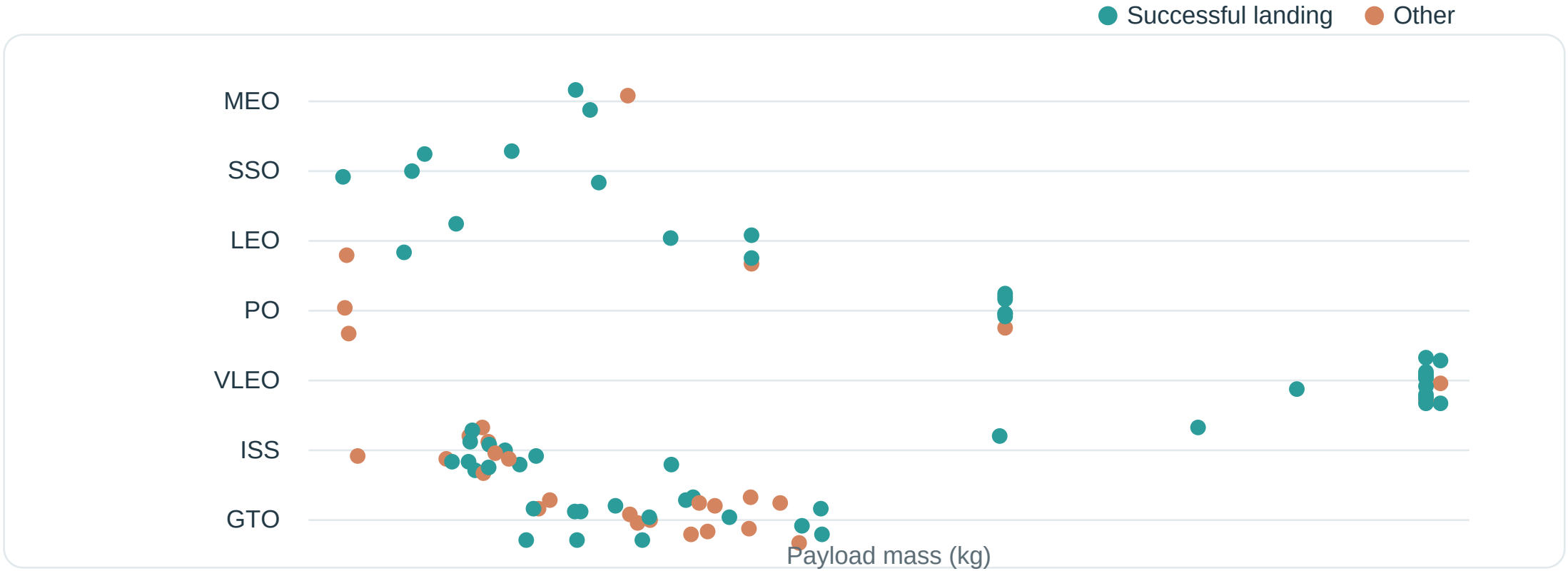
Flight number and orbit

The mix of missions changes across the 2010-2020 flight sequence.



Payload mass and orbit

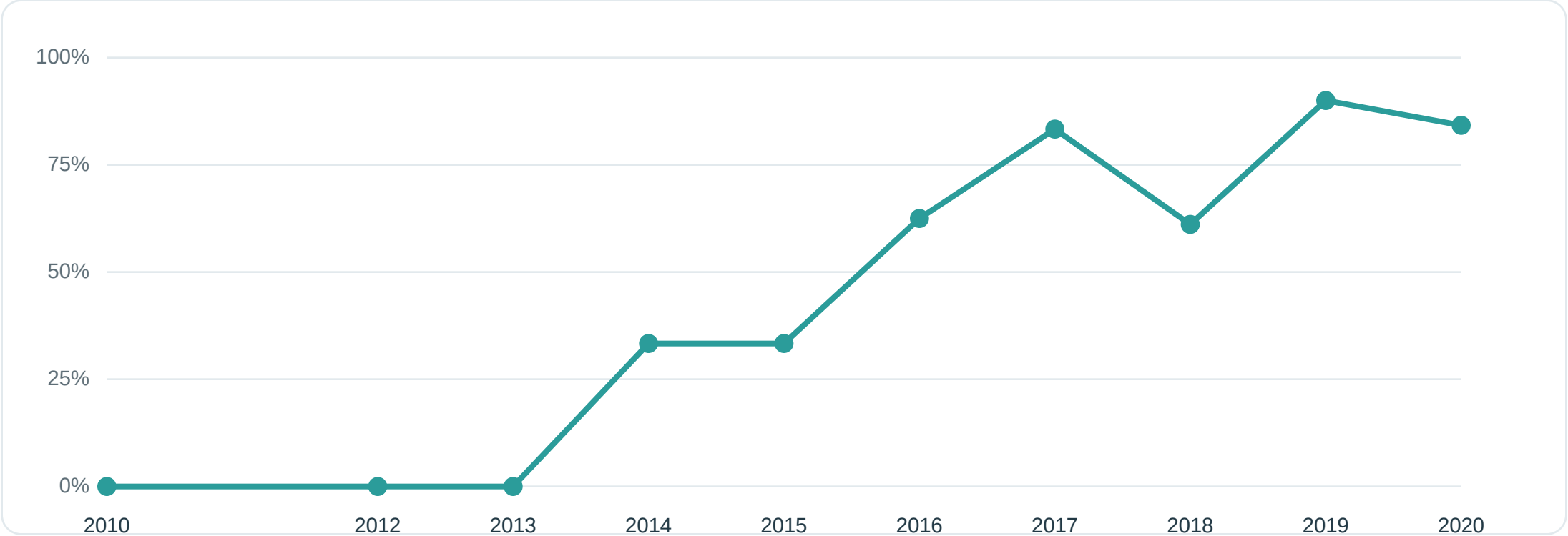
GTO and VLEO include many of the heavier payloads in this snapshot.



EDA RESULTS

Yearly landing trend

The annual mean rises from zero in the earliest years, with variation after 2017.



SQL RESULTS

All launch site names

Four unique SQL site labels; two CCAFS strings refer to variants in the teaching data.

```
SELECT DISTINCT Launch_Site FROM SPACEXTBL ORDER BY Launch_Site
```

Result (4 rows)

Launch_Site: CCAFS LC-40

Launch_Site: CCAFS SLC-40

Launch_Site: KSC LC-39A

Launch_Site: VAFB SLC-4E

SQL RESULTS

First five CCA records

The first five rows matching the CCA prefix start in June 2010.

```
SELECT Date, Launch_Site FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

Result (5 rows)

Date: 2010-06-04 | Launch_Site: CCAFS LC-40

Date: 2010-12-08 | Launch_Site: CCAFS LC-40

Date: 2012-05-22 | Launch_Site: CCAFS LC-40

Date: 2012-10-08 | Launch_Site: CCAFS LC-40

Date: 2013-03-01 | Launch_Site: CCAFS LC-40

SQL RESULTS

Total NASA payload mass

The query sums all records whose Customer field contains NASA.

```
SELECT SUM(PAYLOAD_MASS__KG_) AS total_kg FROM SPACEXTBL WHERE Customer LIKE '%NASA%'
```

Result (1 row)

total_kg: 107010

SQL RESULTS

F9 v1.1 average payload

The mean uses booster labels beginning with F9 v1.1.

```
SELECT AVG(PAYLOAD_MASS__KG_) AS average_kg FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1%'
```

Result (1 row)

average_kg: 2534.6666666666665

SQL RESULTS

First ground-pad landing

The earliest successful ground-pad landing in the SQL snapshot is 2015-12-22.

```
SELECT MIN(Date) AS first_date FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)'
```

Result (1 row)

first_date: 2015-12-22

SQL RESULTS

Drone-ship landings, 4-6 t

Four matching boosters carried payloads above 4,000 kg and below 6,000 kg.

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND
PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000
```

Result (4 rows)

Booster_Version: F9 FT B1022 | PAYLOAD_MASS__KG_: 4696

Booster_Version: F9 FT B1026 | PAYLOAD_MASS__KG_: 4600

Booster_Version: F9 FT B1021.2 | PAYLOAD_MASS__KG_: 5300

Booster_Version: F9 FT B1031.2 | PAYLOAD_MASS__KG_: 5200

SQL RESULTS

Mission outcomes

Landing success and mission success are different targets.

```
SELECT Mission_Outcome, COUNT(*) AS launches FROM SPACEXTBL GROUP BY Mission_Outcome ORDER BY launches DESC
```

Result (4 rows)

Mission_Outcome: Success | launches: 98

Mission_Outcome: Success (payload status unclear) | launches: 1

Mission_Outcome: Success | launches: 1

Mission_Outcome: Failure (in flight) | launches: 1

SQL RESULTS

Maximum payload boosters

Several boosters tie at the dataset maximum of 15,600 kg.

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT
MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)
```

Result (12 rows)

Booster_Version: F9 B5 B1048.4 | PAYLOAD_MASS__KG_: 15600

Booster_Version: F9 B5 B1049.4 | PAYLOAD_MASS__KG_: 15600

Booster_Version: F9 B5 B1051.3 | PAYLOAD_MASS__KG_: 15600

Booster_Version: F9 B5 B1056.4 | PAYLOAD_MASS__KG_: 15600

Booster_Version: F9 B5 B1048.5 | PAYLOAD_MASS__KG_: 15600

Booster_Version: F9 B5 B1051.4 | PAYLOAD_MASS__KG_: 15600

Booster_Version: F9 B5 B1049.5 | PAYLOAD_MASS__KG_: 15600

Booster_Version: F9 B5 B1060.2 | PAYLOAD_MASS__KG_: 15600

... and 4 more matching rows.

SQL RESULTS

2015 drone-ship failures

Two records match the failure-on-drone-ship outcome in 2015.

```
SELECT Date, Booster_Version, Launch_Site, Landing_Outcome FROM SPACEXTBL WHERE Date LIKE '2015%' AND
Landing_Outcome LIKE 'Failure (drone ship)'
```

Result (2 rows)

Date: 2015-01-10 | Booster_Version: F9 v1.1 B1012 | Launch_Site: CCAFS LC-40 | Landing_Outcome: Fa...

Date: 2015-04-14 | Booster_Version: F9 v1.1 B1015 | Launch_Site: CCAFS LC-40 | Landing_Outcome: Fa...

SQL RESULTS

Landing outcome ranking

No attempt is the largest single category in the date window.

```
SELECT Landing_Outcome, COUNT(*) AS launches FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY launches DESC
```

Result (8 rows)

Landing_Outcome: No attempt | launches: 10

Landing_Outcome: Success (drone ship) | launches: 5

Landing_Outcome: Failure (drone ship) | launches: 5

Landing_Outcome: Success (ground pad) | launches: 3

Landing_Outcome: Controlled (ocean) | launches: 3

Landing_Outcome: Uncontrolled (ocean) | launches: 2

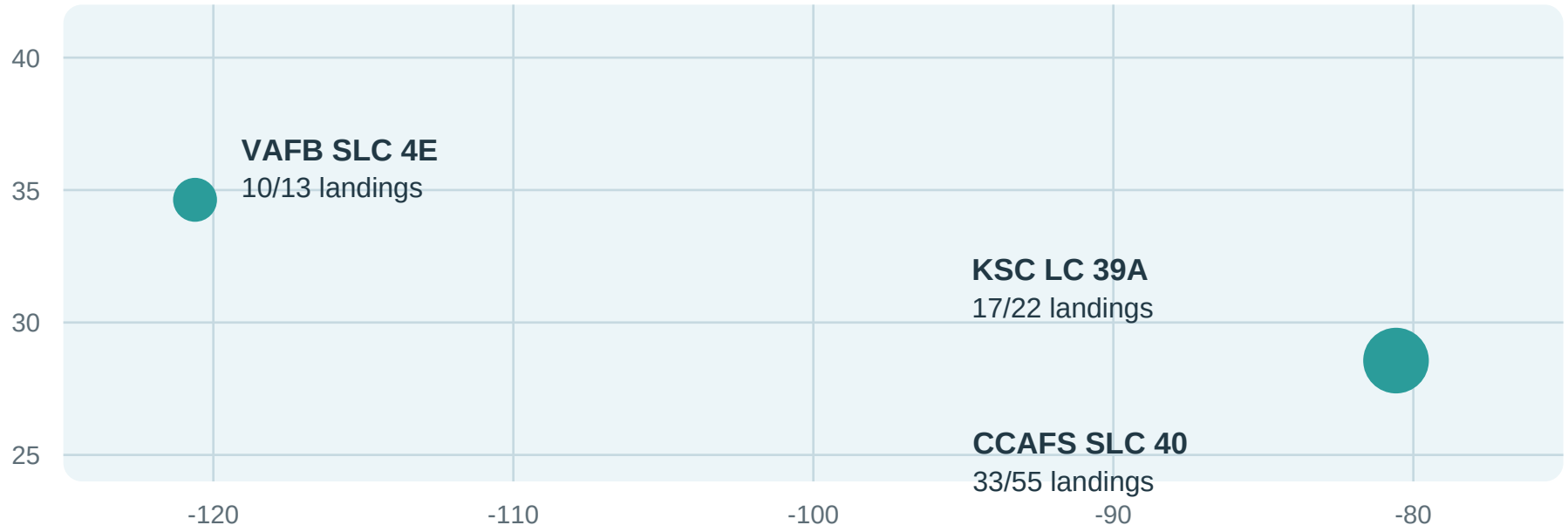
Landing_Outcome: Failure (parachute) | launches: 2

Landing_Outcome: Precluded (drone ship) | launches: 1

FOLIUM MAP RESULTS

Launch site coordinates

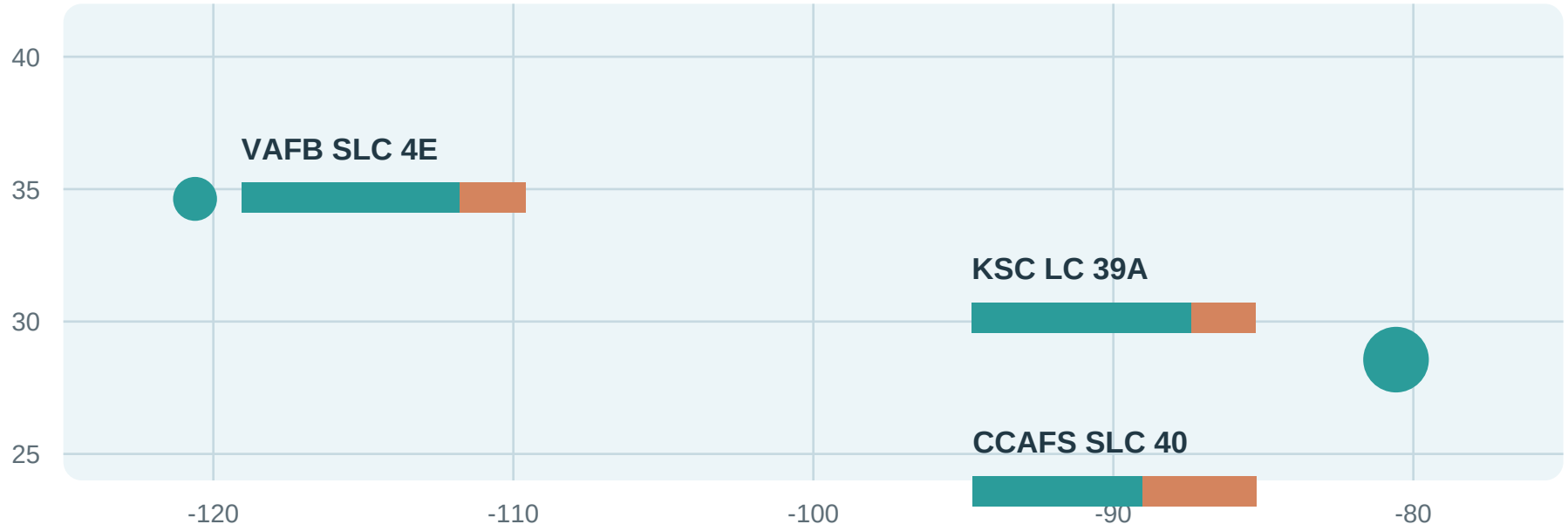
Markers in the interactive Folium map show the three sites in California and Florida.



FOLIUM MAP RESULTS

Landing outcomes by site

Marker popups report launch counts and successful first-stage landings.

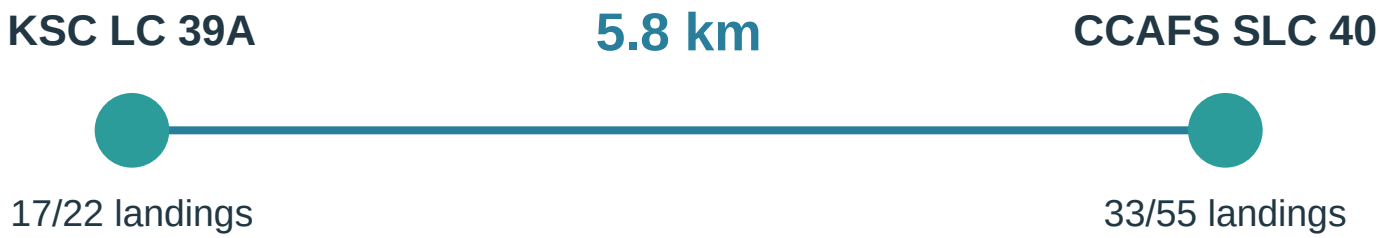


KSC LC 39A: 17/22 (77.3%) | VAFB SLC 4E: 10/13 (76.9%) | CCAFS SLC 40: 33/55 (60.0%)

FOLIUM MAP RESULTS

Florida site proximity

The two Florida launch sites are close together; this is not evidence of a causal site effect.

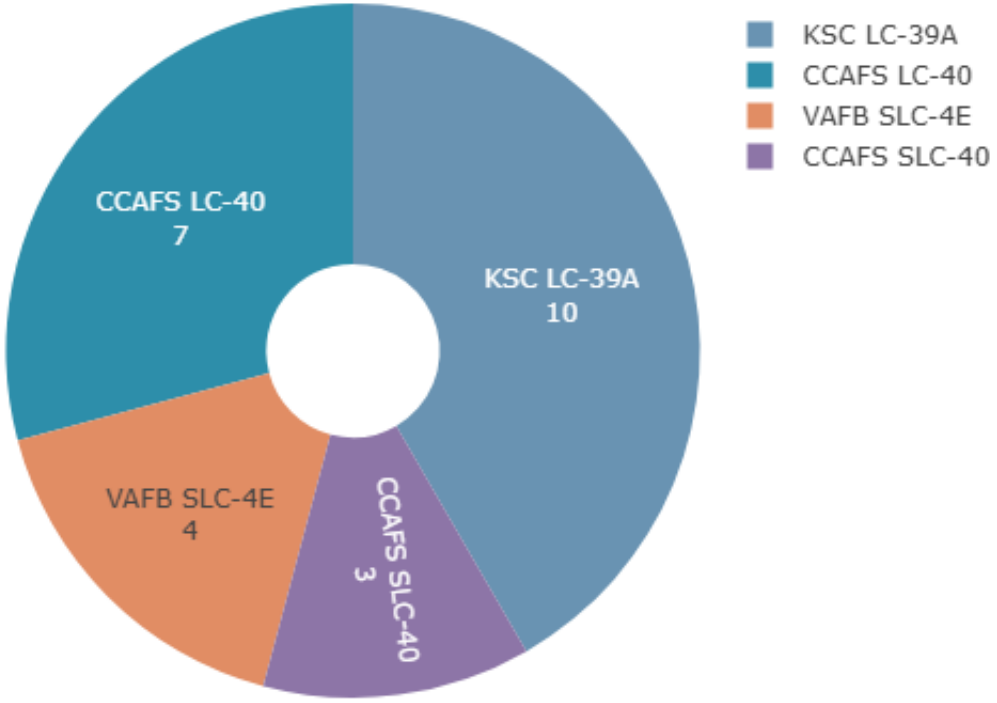


Great-circle distance between the coordinates in the IBM dataset. The interactive Folium HTML also shows the sites on an OpenStreetMap base map.

PLOTLY DASHBOARD RESULTS

Successful landings by site

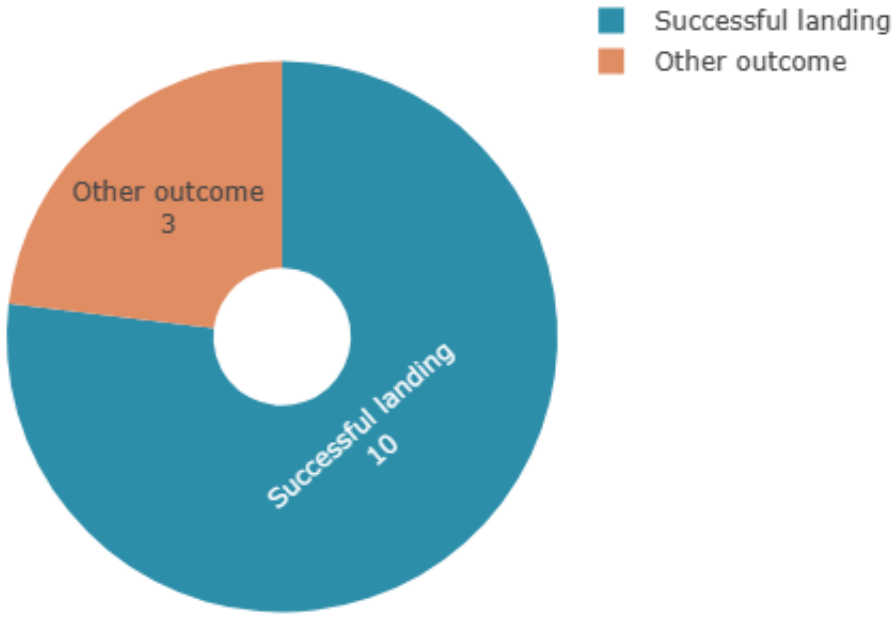
The 56-row dashboard snapshot has



PLOTLY DASHBOARD RESULTS

KSC LC-39A outcomes

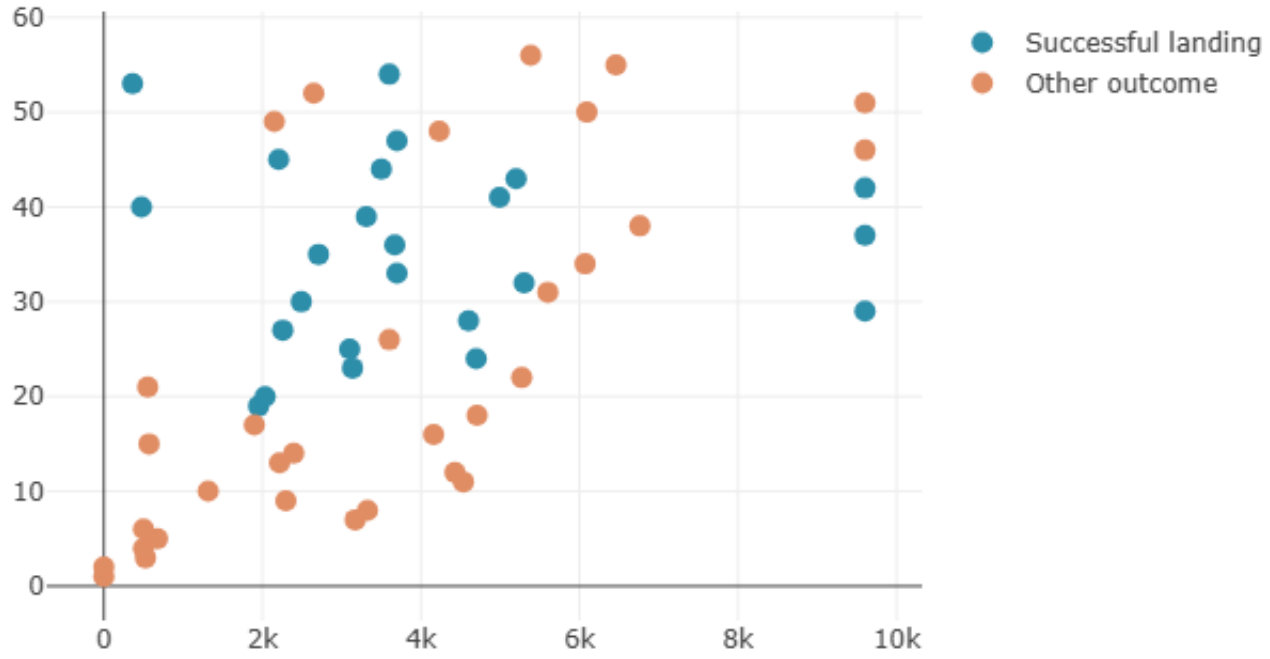
Selecting KSC LC-39A shows 10 succes



PLOTLY DASHBOARD RESULTS

Payload and launch outcome

The dashboard's range con



PREDICTIVE ANALYSIS RESULTS

Classification model accuracy

The model choice uses cross-validation on the 72 training launches.



Selected: decision tree, max depth 2. Mean CV accuracy 83.4%; independent test accuracy 77.8% (14 of 18).

Decision tree confusion matrix

Rows are actual outcomes, columns are predictions on the untouched test split.

	Predicted: other	landing
Actual: other	6	0
Actual: landing	4	8

The tree correctly classifies all 6 non-landings but misses 4 of 12 landings in this small test split.

TAKEAWAYS

Conclusions

- The historical first-stage landing rate is 66.7% across 90 launches. It improves markedly after the earliest years, but the annual groups vary in size.
- Location and mission type are associated with different outcomes. Because the launch mix changes over time, these descriptive differences cannot identify a site or orbit as a cause.
- The selected shallow tree exceeds the majority-class baseline on this split (77.8% versus 66.7%). A larger, more recent dataset and repeated external validation would be needed before operational use.

APPENDIX

Data and reproduction notes

- IBM Skills Network: datasets/dataset_part_2.csv and datasets/dataset_part_3.csv
- IBM Skills Network: labs/module_2/data/Spacex.csv
- IBM Skills Network: datasets/spacex_launch_dash.csv
- Local source files: analyze.py, visuals.py, launch_sites_map.html and launch_dashboard.html

Source base: https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/

The report describes a historical course snapshot. Model estimates are limited by small sample size and possible changes in launch operations after 2020.